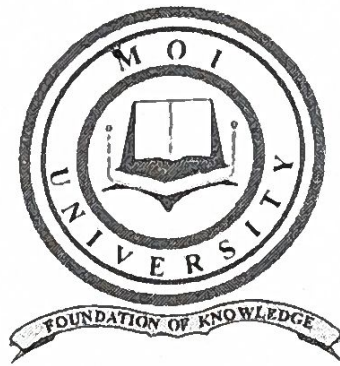




MUEO

MOI UNIVERSITY

OFFICE OF THE DEPUTY VICE CHANCELLOR
(ACADEMICS, RESEARCH, EXTENSION & STUDENT AFFAIRS)

UNIVERSITY EXAMINATIONS

2025/2026 ACADEMIC YEAR

^{FOURTH}
THIRD YEAR FIRST SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR

OF SCIENCE IN

HUMAN RESOURCE MANAGEMENT

COURSE CODE: BBM 354 / *BBM 452*

COURSE TITLE: SOFTWARE DEVELOPMENT I

DATE: TUESDAY- 20TH JANUARY, 2026 TIME: 2:00 P.M. - 5:00 P.M.

INSTRUCTION TO CANDIDATES

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11 February 2026 10:45

BBM354: SOFTWARE DEVELOPMENT

SECTION ONE – COMPULSORY (24 MARKS)

QUESTION ONE

- Explain why software development matters today. (3mks)
- Identify five key skills needed by a software developer (5mks)
- A software project failed due to unclear requirements. Apply SDLC Phase I activities to explain how this failure could have been prevented. *Identifying stakeholder - technical environment* (4mks)
- An organization wants to protect employees salary data in its payroll system. Explain how encapsulation in OOP supports data security. (4mks)
- A system is slow when many users log in simultaneously. Identify one non-functional testing technique that would address this issue and explain why *Performance testing* (4mks)
- Explain how the V-Model improves system reliability and safety. *- supports validation - reliability - supports verification - safety* (4mks)

SECTION TWO – ANSWER ANY THREE QUESTIONS

(36 MARKS)

QUESTION TWO

(12marks)

A manufacturing company plans to develop an Inventory Management System.

- Explain four key phases of the Software Development Life Cycle (SDLC) and how each phase supports successful system development. *- Gathering Req's - Maintenance - Design - Deployment and testing - data migration and setting up product - releasing system* (8mks)
- Analyze two risks the company may face if SDLC phases are not properly followed. *- security issues* (4mks)

QUESTION THREE

(12marks)

- A development team is working remotely on a payroll system. Explain how Git supports collaboration, accountability, and version control. *Method, balance, directly used* (6mks)
- Explain how inheritance and polymorphism improve maintainability. *not class, device properties, flexibility, code extraction* (6mks)

QUESTION FOUR

(12marks)

An organization is migrating from a monolithic system to microservices.

- Analyze the limitations of monolithic architecture in a growing business. *- Entire up as a single unified unit - easier to develop - Scalability challenges*
- Explain two benefits of microservices architecture in modern enterprises. *- Smaller, independent services - Flexible to scale - Complex to manage many products*

QUESTION FIVE

(12marks)

A startup firm adopts Agile methodologies to develop a web application intended for customer use.

- Analyze how Agile methodologies improve flexibility and customer satisfaction (6mks)
- Evaluate the impact of cloud computing and IoT on modern business software systems. (6mks)